

NeoStats Credit Risk Intelligence Platform

Calibrated EBM | PostgreSQL analytics | Gemini agentic assistant

Demo-ready credit-risk decision support

Transparent prediction, model explanations, curated analytics,
and durable conversational intelligence.

Internal risk segmentation - not an automated lending policy or regulatory rating.

1. Confirmed EDA

Associations measured across 307,511 applicants

Portfolio

8.07% default rate
11.4:1 class imbalance
Evaluate with ROC-AUC, PR-AUC, and calibration

Segments

20-29 has the highest observed age-band rate
Rented housing shows 1.53x observed lift
Associations are not causal policy rules

Behaviour

External scores dominate importance
Overdues, refusals, and late instalments provide strong historical signals

2. Model hierarchy

Untouched stratified test-set results

Logistic baseline

ROC-AUC 0.759

PR-AUC 0.238

Brier 0.199

Interpretable performance floor

LightGBM shadow

ROC-AUC 0.778

PR-AUC 0.265

Brier 0.174

Feature pruning and reference
ceiling

Calibrated EBM

ROC-AUC 0.764

PR-AUC 0.250

Brier 0.068

Production glass-box model

3. Explainability and rules

Transparent evidence with explicit governance boundaries

EBM explanations

Exact additive global and applicant-level score contributions support model review. Customer-facing reasons require separate validation.

Surrogate tree

Depth 4
Validation R2 0.525
Risk-band agreement 68.0%
Explanatory approximation only - not approval policy.

4. Agentic assistant

Fast routing, bounded execution, and transparent evidence

Database

Curated PostgreSQL views
Read-only login
8-second timeout
200-row maximum

Knowledge

206 curated chunks
Local 768-dimensional embeddings
pgvector cosine retrieval
Confirmed internal evidence

Web

DDGS external search
Maximum five results
Internal-data guardrail
Transparent offline fallback

5. Deployment and verification

One-command Docker Compose demo

Runtime

PostgreSQL 17 + pgvector
One-shot deterministic initialization
Streamlit waits for initialization
Models and artifacts mounted read-only

Verification

33 automated tests
Persistent refresh-safe chat threads
Positive SQL function allowlist
Healthy Streamlit endpoint